

2009 F=ma Exam: Problem 10

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From kinematics, the trajectory of the first apple is

$$y_1(t) = vt - \frac{1}{2}gt^2$$

The trajectory of the second apple is

$$y_2(t) = -vt - \frac{1}{2}gt^2$$

The displacement between the two apples is then

$$\Delta y(t) = y_1(t) - y_2(t) = 2vt$$

At $t = 2$ s, we have

$$\Delta y = 2(7 \text{ m/s})(2 \text{ s}) = 28 \text{ m}$$

so the answer is C.